KENYA MEDICAL TRAINING COLLEGE
FACULTY OF REHABILITATIVE SCIENCES
DEPARTMENT OF MEDICAL ENGINEERING
FINAL QUALIFYING EXAMINATION
FOR
CERTIFICATE IN MEDICAL ENGINEERING TECHNOLOGY

PAPER: ELECTRICAL ENGINEERING TECHNOLOGY

DATE: 14th February 2020	**TIME: 9:00AM – 12.00NOON**

INSTRUCTIONS

1. This paper consists of:
   - Section 1 (40 Multiple Choice Questions)
   - Section 2 (8 Short Answer Questions)
   - Section 3 (1 Long Answer Question)

2. Attempt **ALL** Questions

3. Write the **EXAMINATION NUMBER** given on all the answer sheets provided and on the question paper.

4. Ensure that all examination answer scripts are handed in at the end of the examination

5. Ensure you sign the examination register provided

EXAMINATION NUMBER .....

SECTION 1
MULTIPLE CHOICE QUESTIONS (40 MARKS)

1. The greater the number of turns of coil
   a. The smaller will be the strength of the magnetic field
   b. The greater will be the strength of magnetic field
   c. The smaller will be the strength of the electric field
   d. The greater will be the strength of the electric field
2. What is the disadvantage of using a short-pitched coil
   a. Total voltage around the coils will be reduced
   b. Losses are reduced
   c. Efficiency increases
   d. Total voltage around the coil is increased
3. A one winding consists of how many coils and contractors
   a. Two, one
   b. One, one
   c. One, two
   d. Half, one
4. For a 2-pole machine, 360^0 of electrical cycle is equal to how many degree of mechanical operation
   a. 180^0
   b. 360^0
   c. 90^0
   d. 270^0
5. Which law is used in computation of developed E.M.F.
   a. Farady's law
   b. Amperes law
   c. Maxwells law
   d. Biot saxart's law
6. The process of producing magnetic field in a system is known as
   a. Induced EMF
   b. Excitation
   c. Experienced force
   d. Torque development
7. Where is the field winding mounted in a DC machine
   a. Stator
   b. Rotor
   c. Commutator
   d. Brushes

8. What is the material used for the brushes in DC machines
   a. Iron
   b. Carbon
   c. Aluminum
   d. Steel
9. In a DC machine, rectification process is created out in order to get unidirectional output (DC). The rectification process is carried out by
   a. Half way rectifier
   b. Full wave rectifier
   c. Mechanical rectifier composed of commutator brushes assembly
   d. Centre tapped rectifier
10. DC shunt motors are commonly used in
   a. Cranes
   b. Electric traction
   c. Elevator
   d. Ladle machines
11. When a motor runs on no load, then
   a. Back EMF is almost equal to the applied voltage
   b. Back EMF will be greater than the applied voltage
   c. Back EMF will be less than applied voltage
   d. Back EMF is exactly equal to the applied voltage
12. In the biomedical instruments like the artificial heart pumps, the commonly used motor is
   a. Permanent magnet DC motor
   b. Brushless DC motor
   c. Shunt motor
   d. Series motor
13. Due to the effect of armature reaction in a DC machine, the value of generated voltage and flux per pole
   a. Increase, decrease
   b. Decrease, decrease
   c. Decrease, increase
   d. Increase, increase
14. When the speed of an alternator is increased, the frequency
   a. Increases
   b. Decreases
   c. Remains constant
   d. May increase or decrease depending on power factor

15. A generator which gives DC supply to the rotor of an alternator is called
   a. Converter
   b. Exeter
   c. Inverter
   d. Rectifier
16. The advantage of salient pole in an alternator is
   a. Reduce windage losses
   b. Reduce noise
   c. Reduce bearing loads and noise
   d. Adaptability to low and medium operation span
17. The EMF generated in an alternator depends on
   a. Frequency
   b. Flux per pole
   c. Coil span factor
   d. All of the above
18. The exciter for an AC generator is
   a. Shunt motor
   b. Serra motor
   c. Shunt generator
   d. Compound generator
19. Alternators with a cylindrical pole type of rotors are generally used with prime movers of
   a. High speed
   b. Low speed
   c. Medium speed
   d. Both low and high speed
20. The frequency of voltage generated by an alternator having 8 poles and rotating at 250RPM
   a. 60Hz
   b. 50Hz
   c. 25Hz
   d. 6 2/3 Hz
21. In order that two alternators be operated in parallel, which of the following factors shold be identical
   a. Voltage
   b. Frequency
   c. Phase sequence
   d. All of the above

22. The armature winding of an alternator is
   a. Star-delta connected
   b. Generally delta connected
   c. Always star connected
   d. Delta star connected
23. In a synchronous motor, the additional damper winding is mounted on
   a. Rotor
   b. Stator
   c. Field winding
   d. All of the above
24. A damper winding in a synchronous motor is
   a. Used for starting of synchronous motor
   b. Used to increase the stability of the motor during load transient
   c. A group of copper bards short circuited at the end
   d. All of the above
25. A maximum torque which a synchronous motor can develop at rated voltage and frequency without losing synchronism is called
   a. Running torque
   b. Locked rotor torque
   c. Pull-in torque
   d. Pull-out torque
26. An induction motor can be said to be analogous to
   a. Transformer
   b. Synchronous motor
   c. Universal motor
   d. Stepper motor
27. A three (3) phase induction motor with rotor circuit open will
   a. Not run
   b. Run normally
   c. Get overheated
   d. Make noise
28. The rotor of an induction motor must never run at a synchronous speed because the relative speed between the rotating flux and rotor will be
   a. Zero and hence torque zero
   b. Zero and hence torque maximum
   c. Maximum and have zero torque
   d. Maximum and torque will be zero

29. A synchronous motor which works on leading power factor and does not drive a mechanical load is called
   a. Static condenser
   b. Condenser
   c. Synchronous condenser
   d. All of the above
30. During hunting of a synchronous motor
   a. Negative phase sequence currents are generated
   b. Harmonics are developed in the armature circuits
   c. Damper bars develop torque
   d. Field excitation increases
31. Inserting of external resistance in the squirrel cage induction motor
   a. Increases torque
   b. Decreases torque
   c. Increases torque for some seconds then decreases
   d. Does not affect
32. Starters are used in induction motors because
   a. Starting torque is high
   b. It is run against heavy load
   c. It cannot run in reverse direction
   d. Its starting current is 5 times or more of rated current
33. D0L starters are strictly restricted to induction motors with
   a. Low-horse power
   b. Variable speed motor
   c. High horsepower motors
   d. High speed motors
34. Which of the following has the highest starting toque
   a. Universal motor
   b. Capacitor start
   c. Capacitor start capacitor run
   d. All have zero starting torque
35. Which one of the following motors are used in a household refrigerator
   a. DC series motor
   b. DC short motor
   c. Universal motor
   d. Single phase induction motor
36. Slip speed is
   a. Difference in synchronous speed and actual rotor speed
   b. Difference of actual tor speed and synchronous speed
   c. Sum of synchronous and rotor speed
   d. Half the sum of synchronous rotor speed

37. A transformer works on the principle of
   a. Self-induction
   b. Mutual induction
   c. Faradays Law of electromagnetic induction
   d. Self and mutual induction

38. A transformer transforms
   a. Frequency
   b. Only voltage
   c. Only current
   d. Both current and voltage

39. Transformer oil is used to provide
   a. Insulation and cooling
   b. Cooling and lubrication
   c. Lubrication and insulation
   d. Insulation cooling and lubrication

40. Eddy currents losses in a transformer core may be reduced by
   a. Reducing air gap in the magnetic current
   b. Reducing the thickness of lamination
   c. Increasing the thickness of lamination
   d. Increasing the gap in the magnetic current

SECTION 2
SHORT ANSWER QUESTIONS (40 MARKS)

1. A series circuit having a resistance of 100Ω connected with 50μF and 0.5H inductive coil across 220V 50Hz supply. Calculate;
   (i) Impedance (z) **(2 marks)**
   (ii) Current (I) **(1 mark)**
   (iii) Power factor **(1 mark)**
   (iv) Real power **(1 mark)**

2. A balanced 3-phase start connected load draws a current of 25A at lagging power factor of 0.8 when connected to a three-phase supply 400V line-line. Calculate
   (i) Phase voltage **(1 mark)**
   (ii) Real power **(2 marks)**
   (iii) Reactive power **(2 marks)**

3. A DC generator produces an armature current of 50A when generating 400V. If the terminal voltage is 390V.
   Calculate;
   (i) Value of armature resistance **(3 marks)**
   (ii) Power loss in the armature circuit **(2 marks)**

4. (a) Explain why E.M.F. generated in the armature of a DC motor is called back E.M.F. **(2 marks)**

(b) The power input to a 230V DC shunt motor is 8.477KW **(3 marks)**
The field resistance is 230Ω and the armatureresistance is 0.28Ω .
Find;
(i) Input current **(1 mark)**
(ii) Armature current **(1 mark)**
(iii) Back EMF **(1 mark)**

5. Compare and contrast the salient pole and non-salient pole type of rotors in an AC generator **(5 marks)**

6. Explain the following
(i) How does a single phase induction motor differ from a three-phase induction motor **(2 marks)**
(ii) Why single phase induction motors are not self-starting **(2 marks)**
(iii) Which type of single phase induction motor has the lowest starting torque **(1 mark)**

7. (a) Why do large inductions motors with power 5HP and above require a suitable starting method. **(2 marks)**
(b) List the various starting methods for starting induction motors **(3 marks)**

8. (a) Explain why a transformer is the most efficient electrical machine **(2marks)**
(b) An ideal transformer connected to a 240V main supplies a 12V 150W lamp. Calculate the transformer's turns ratio and current taken form the supply. **(3 marks)**

SECTION 3
LONG ANSWER QUESTIONS (20 MARKS)

1. (a) Clearly outline and explain the types of energy losses that occur in a DC machine **(6 marks)**
(b) A series wound DC motor drives a load at 30revs and takes a current of 10A, when the supply voltage is 400V. If the total resistance of the motor is 2Ω and the iron friction and windage losses amount 300W. Determine the efficiency of the motor. **(4 marks)**
(c) Explain the reason why rotating field with a stationary armature design is preferred in large alternators **(4 marks)**
(d) As an alternator is loaded, its terminal voltage is found to vary. List three reasons for the variation in terminal voltage. **(3 marks)**
(e) The effective resistance of 2200V, 50Hz 440KVA 1 phase alternator is 0.5Ω. On short circuit, current of 40A gives a full load current of 200A. The EMF on open-circuit with the same field excitations is 1160V. Calculate the synchronous impedance and reactance **(3 marks)**